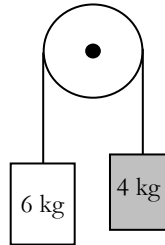


CONNECTED PARTICLES: PULLEYS

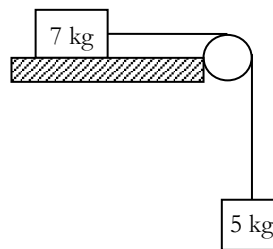
03/11/25

Draw force diagrams for each of the following situations

- Two weights, of mass 6 kg and 4 kg respectively, are connected by a light inextensible string running over a smooth pulley.



- Copy the diagram and mark on the forces acting on the weights.
 - Find the acceleration of the two weights and also the force the string exerts on them.
- Two weights, of mass 7 kg and 5 kg respectively, are connected by a light inextensible string running over a smooth pulley as shown in the diagram. The 7 kg mass lies on a smooth horizontal table and the other mass hangs freely. Find the acceleration of the objects and hence find the tension in the string.



- Three weights, of masses 7 kg, 10 kg and 4 kg, are connected by light inextensible strings running over smooth pulleys as shown in the diagram. The 10 kg mass lies on a smooth horizontal table and the other two masses hang freely. Find the acceleration of the objects and hence find the tension in both strings.

